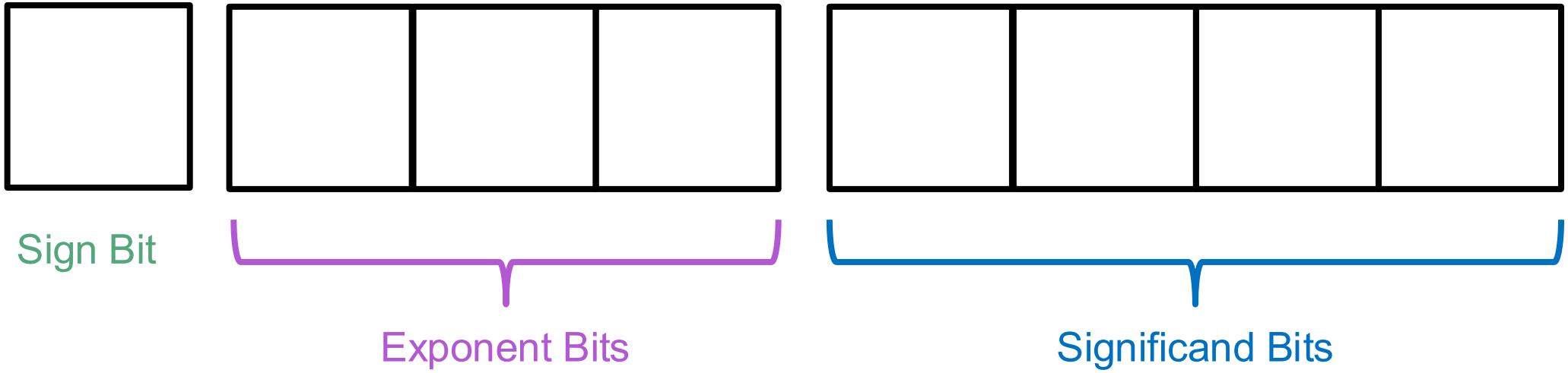


Minifloat

COMP211 Connor McMahon

Minifloat



Sign Bit (1 bit):
- 0: Positive number
- 1: Negative number

Exponent (3 bits):
- Represents the exponent using a fixed **bias of 3**

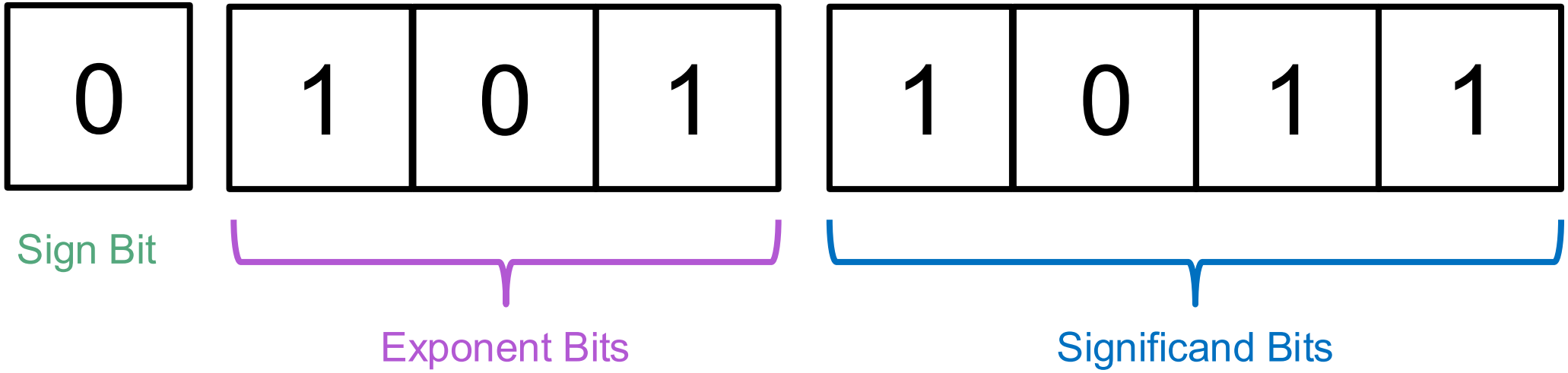
Significand (4 bits):
- Represents the fractional part
- Assumes an **implicit leading 1** for normalized numbers

Bias

- Why do we use a bias instead of 2's complement?
- Simplifies Comparison Operations
  - With a biased exponent, larger numbers have higher bit values, and smaller numbers have lower bit values, making it easy to compare them as unsigned integers.
  - If two's complement were used, negative exponents (which have leading 1s) would be greater in unsigned comparison, breaking the ordering.

$$\text{Bias} = 2^{n-1}-1$$

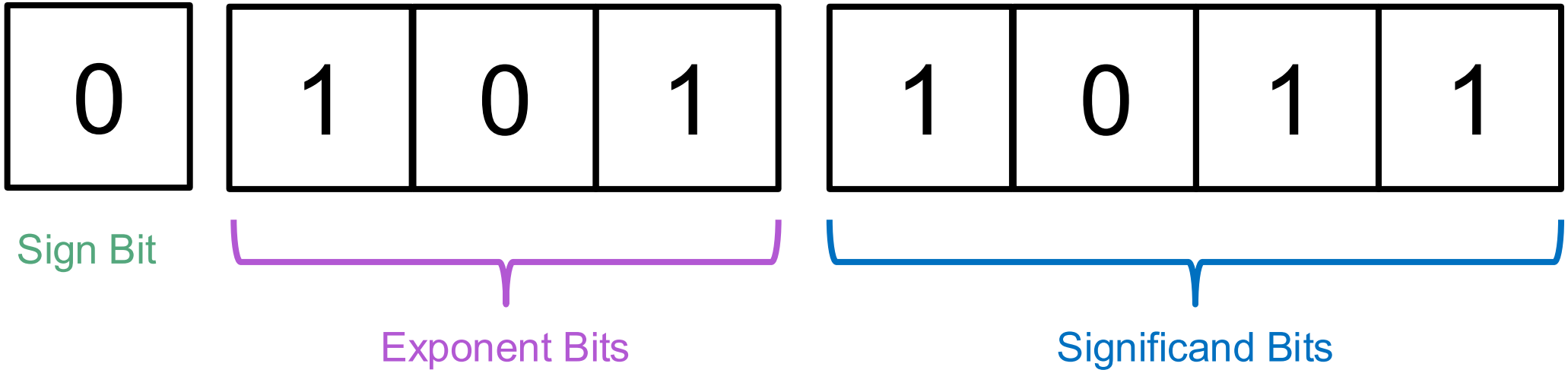
Minifloat



Bias = 3

$(-1)^{S} \times 1.\langle\text{significand bits}\rangle \times 2^{\langle\text{exponent bits}\rangle - \langle\text{bias}\rangle}$

Minifloat



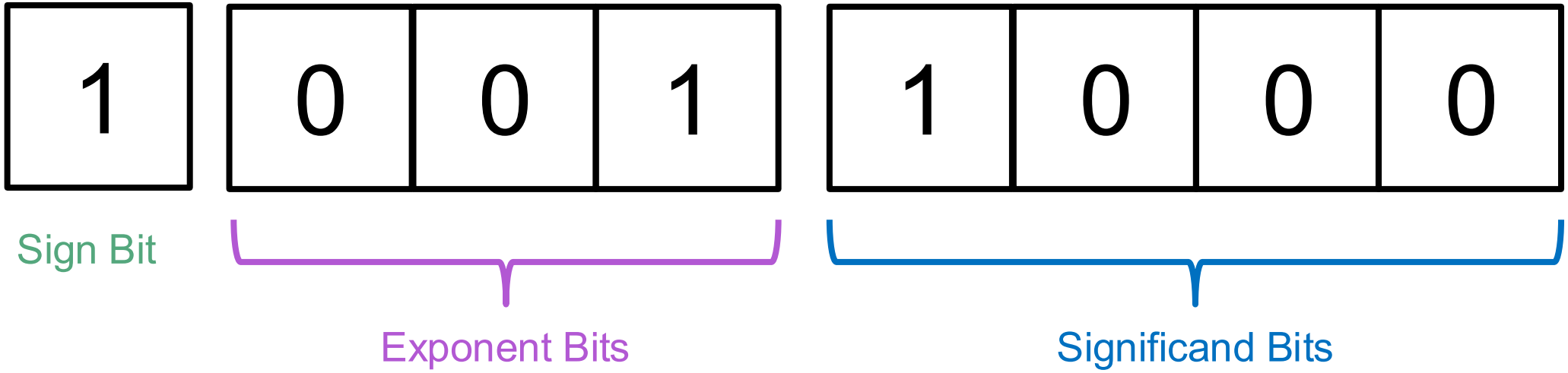
Bias = 3

$$(-1)^{S} \times 1.\langle \text{significand bits} \rangle \times 2^{\langle \text{exponent bits} \rangle - \langle \text{bias} \rangle}$$

$$(-1)^{0} \times 1.1011 \times 2^{5-3}$$

$$110.11 = 6.75$$

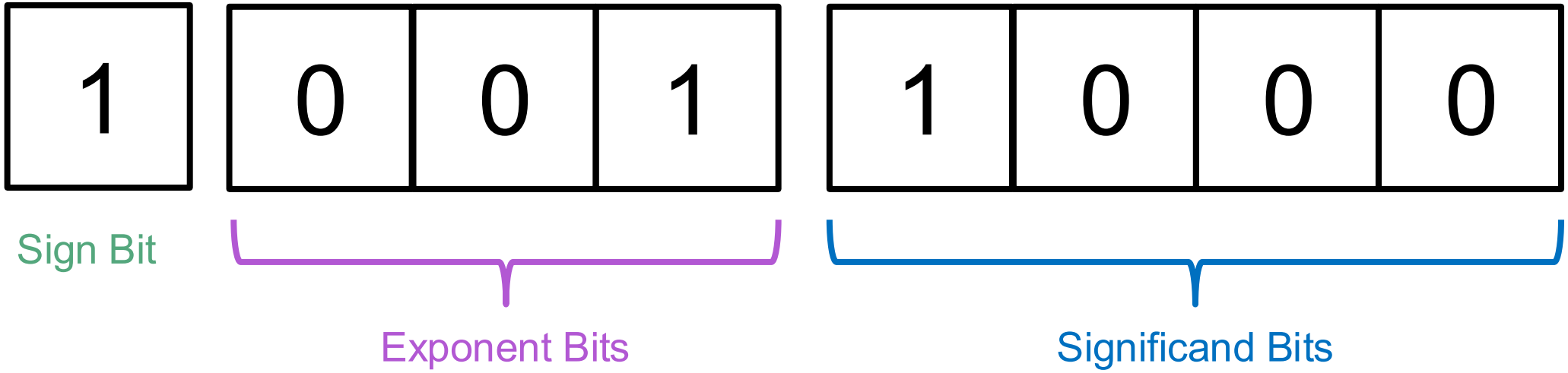
Minifloat



Bias = 3

$(-1)^{S} \times 1.\langle\text{significand bits}\rangle \times 2^{\langle\text{exponent bits}\rangle - \langle\text{bias}\rangle}$

Minifloat



Bias = 3

$$(-1)^{S} \times 1.\langle\text{significand bits}\rangle \times 2^{\langle\text{exponent bits}\rangle - \langle\text{bias}\rangle}$$

$$(-1)^{1} \times 1.1000 \times 2^{1-3}$$

$$-0.011000 = -0.375$$

Minifloat

- For each value below, determine if it can be represented.
  - If it can be represented, write out its Minifloat encoding.
  - If not, explain why: not enough precision or not enough range.
- A) 15
- B) -4.25
- C) -32
- D) 12.25

